Opgave: Dag 7 eksempel 7.4
(ELE1)

Frederik Rasmussen

VIA University College

Del 1 - RMS-fasorer

Tallene foran cosinus er amplituder. Kompleks effekt regnes med RMS-værdier, så vi dividerer med √2:

$$\tilde{V} = V_{\text{rms}} = \frac{V}{\sqrt{2}} \qquad (6.10)$$

Complex Power – Example 7.4

EXAMPLE 7.4 Complex Power Calculations

roblem

se the definition of complex power to calculate real and reactive power for the load of igure 7.12.

olution

nown Quantities: Source, load voltage, and current.

ind: $S = P_{av} + jQ$ for the complex load.

chematics, Diagrams, Circuits, and Given Data: $v(t) = 100\cos(\omega t + 0.262)$ V; $t) = 2\cos(\omega t - 0.262)$ A.

ssumptions: Use rms values for all phasor quantities in the problem.

$$V_t := \frac{100}{\sqrt{2}} \cdot \exp(1\text{j} \cdot 0.262) \cdot V = (68.298 + 18.315\text{j})\ V \qquad I_t := \frac{2}{\sqrt{2}} \cdot \exp(-1\text{j} \cdot 0.262) \cdot A = (1.366 - 0.366\text{j})\ A$$

Del 2 - Kompleks effekt

S er spændingen gange den konjugerede strøm. Mathcad har ingen konj()-funktion med navn, så den konjugerede strøm skrives Re(I) – j·Im(I).

$$\mathbf{S} = \tilde{\mathbf{V}}\tilde{\mathbf{I}}^* \qquad \text{Complex power} \qquad (6.24)$$

$$I_k := \text{Re}(I_t) - 1\text{j} \cdot \text{Im}(I_t) \qquad S_c := V_t \cdot I_k = (86.582 + 50.035\text{j})\ W$$

$$|S_c| = 100\ W \qquad \arg(S_c) = 30.023\ deg$$

Non-Commercial Use Only

Realdelen er den aktive effekt P (W) og imaginærdelen den reaktive effekt Q (var):

$$\mathbf{S} = P_{\text{avg}} + jQ \qquad (6.26)$$

$P_a := \mathrm{Re}\left(S_c\right) = 86.582\ W$ $Q_r := \mathrm{Im}\left(S_c\right) = 50.035\ W$ $pf := \cos\left(\arg\left(S_c\right)\right) = 0.866$

Q > 0, så belastningen er induktiv (haltende). Uden √2 ville man have fået det dobbelte.

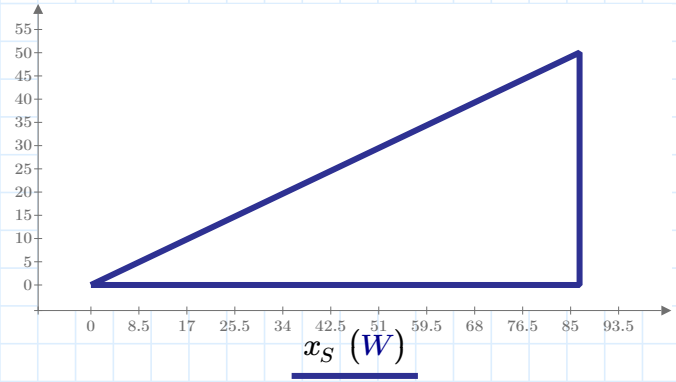
$P_a = 86.582\ W$ $Q_r = 50.035\ W$ $pf = 0.866$

Non-Commercial Use Only

Plot - Effekttrekanten

$$x_S := \begin{bmatrix} 0 \\ P_a \\ P_a \\ 0 \end{bmatrix} \qquad y_S := \begin{bmatrix} 0 \\ 0 \\ Q_r \\ 0 \end{bmatrix}$$

Vandret P = 86,6 W, lodret Q = 50 var, hypotenusen |S| = 100 VA ved 30°.



Non-Commercial Use Only